

Veteran Housing Benefit Eligibility Snapshot Literature Review

Author: Chad King - MS Human Computer Interaction | Jarvis College of Computing and Digital Media | DePaul University

Executive Summary

This literature review surveys peer-reviewed research relevant to the design of the Veteran Housing Benefit Eligibility Snapshot, a structured intake tool that produces a personalized eligibility report for U.S. veterans seeking to identify housing-related benefits they may be entitled to. The Snapshot does not make eligibility determinations; it surfaces likely pathways, next actions, document requirements, and assumptions, and routes users to the authoritative bodies that actually adjudicate eligibility. The design problem is structurally a personalized decision aid problem combined with a benefits take-up problem, and the literature reflects that combination.

Three themes organize the seven sources reviewed. First, research on patient decision aids and AI-assisted decision-making establishes both an evidence base for what makes a personalized decision support tool effective, and an empirical foundation for why AI-generated rationale text must be designed to support deliberation rather than to invite passive acceptance. Second, research on algorithmic eligibility systems documents the harm that automated tools inflict on vulnerable populations when they overclaim authority, while parallel research on benefits take-up documents that the larger problem for most eligible veterans is the opposite, namely never claiming benefits they are entitled to because of informational complexity and low program awareness. Third, research on digital navigation tools and human community health worker navigators offers convergent evidence on what effective orientation toward authoritative pathways looks like for first-time benefits seekers in vulnerable populations. The review concludes with six design implications operationalized in the Snapshot's current output format, and identifies three gaps in the existing literature that the capstone project sits within by design.

Search Methodology

Sources were identified through four databases: ACM Digital Library for HCI venues, the Journal of Medical Internet Research family of journals and the Cochrane Database of Systematic Reviews for decision aid and navigation research, the American Economic Association journals for benefits take-up research, and Google Scholar for cross-disciplinary

coverage. Search terms combined three concept clusters: decision support and AI-generated output ("patient decision aid," "AI-assisted decision-making," "AI explanations," "overreliance," "cognitive forcing"); benefits access and take-up ("benefits take-up," "social benefits," "non-claiming," "informational complexity," "eligibility navigation"); and design or trust constructs ("trust calibration," "navigator," "plain language," "vulnerable populations"). Inclusion criteria were peer-reviewed publication, English language, and direct relevance to at least one of the three themes. Date range was prioritized as 2014 to 2026 for HCI sources to capture work on AI-assisted decision support, with foundational pre-2014 sources included where they remain the canonical reference. The final source set is seven peer-reviewed publications spanning HCI venues (CHI, CSCW), medical informatics venues (Cochrane, JMIR Mental Health, JAMA Internal Medicine), economics (American Economic Review), and one book-length work from a university press.

Theme 1: Personalized Decision Aids and AI-Generated Output for Lay Decision-Makers

The Eligibility Snapshot's output format, in which the user supplies their situation through a structured intake form and receives a personalized report with ranked pathway options, brief rationale, document requirements, and explicit assumptions, places the project within an established research category: the patient decision aid. The decision aid literature is one of the most mature evidence bases in applied informatics and provides both the design vocabulary (options, attributes, values clarification, decisional conflict) and a quality framework (the International Patient Decision Aid Standards) that the Snapshot can be evaluated against. The Snapshot is not a clinical decision aid in domain, but it is one in form. Two further sources address what happens specifically when the rationale section of a decision aid is generated by an AI system: AI-generated explanations can either support a user in calibrating their reliance on the recommendation or, more commonly, can induce passive acceptance of recommendations the user has not engaged with critically.

Stacey, D., Lewis, K. B., Smith, M., Carley, M., Volk, R. J., Douglas, E. E., Pacheco-Brousseau, L., Finderup, J., Gunderson, J., Barry, M., Bennett, C. L., Bravo, P., Steffensen, K. D., Gogovor, A., Graham, I. D., Kelly, S. E., Légaré, F., Sondergaard, H., Thomson, R., Trevena, L., & Witteman, H. O. (2024). Decision aids for people facing health treatment or screening decisions. *Cochrane Database of Systematic Reviews*, 1, CD001431. <https://doi.org/10.1002/14651858.CD001431.pub6>

Stacey and colleagues maintain the long-running Cochrane systematic review of randomized trials of patient decision aids. The 2024 update synthesizes results from over two hundred randomized controlled trials with more than one hundred thousand

participants and finds consistently that decision aids, compared with usual care, improve user knowledge, generate more accurate risk perceptions, increase the user's active role in decision-making, and reduce the proportion of users who remain undecided. The review also catalogues the design attributes that distinguish effective decision aids: focus on a specific decision, presentation of options and their probable outcomes, support for values clarification, and explicit treatment of uncertainty. The relevance to Veteran Housing Benefit Eligibility Snapshot tool is structural rather than topical. The Snapshot's output sections (the BLUF summary, the ranked pathways, the rationale, the assumptions and unknowns, the non-determination disclaimer, and the next actions) correspond almost exactly to the canonical decision aid elements documented in this review. The implication is that the Snapshot can be evaluated against an established quality standard (the International Patient Decision Aid Standards) rather than requiring a bespoke evaluation framework, which is a significant advantage for the capstone's evaluation phase.

Bansal, G., Wu, T., Zhou, J., Fok, R., Nushi, B., Kamar, E., Ribeiro, M. T., & Weld, D. S. (2021). Does the whole exceed its parts? The effect of AI explanations on complementary team performance. In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21). Association for Computing Machinery. <https://doi.org/10.1145/3411764.3445717>

Bansal and colleagues conduct mixed-method user studies testing whether AI-generated explanations improve human-AI team decision-making compared with no explanation. The headline finding is that explanations did not improve team accuracy in the way the literature had assumed; they increased the probability that humans would accept the AI's recommendation, regardless of whether the recommendation was correct. The implication is direct for the Snapshot's Rationale section, which is AI-generated explanation text. A well-written rationale can amplify rather than calibrate trust, particularly when the recommendation itself is wrong. The design response the Veteran Housing Benefit Eligibility Snapshot has adopted is twofold. First, the Rationale section is paired with a separate Assumptions and Unknowns section that explicitly surfaces the brittle inputs to the recommendation, breaking the rationale's fluency with disclosed uncertainty. Second, the Disclaimer section asserts that the Snapshot is not a determination of eligibility, framing the rationale as orientation rather than authoritative inference. These design choices treat the Bansal finding as a constraint to design against, not a result to merely cite.

Buçinca, Z., Malaya, M. B., & Gajos, K. Z. (2021). To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making.

Proceedings of the ACM on Human-Computer Interaction, 5(CSCW1), Article 188.
<https://doi.org/10.1145/3449287>

Buçinca and colleagues build directly on the Bansal finding and ask whether deliberate design interventions can reduce AI overreliance. The theoretical frame is dual-process cognition: users default to System 1 (fast, heuristic, accepting) when reading AI output unless the interface compels them to engage System 2 (slow, deliberative, evaluative). The paper tests three cognitive forcing functions, including requiring the user to commit to a preliminary decision before the AI recommendation is shown, and finds that each intervention reduces overreliance compared with conventional explanation-augmented output. A secondary finding is consequential for the Snapshot's audience: the interventions benefited users with high Need for Cognition substantially more than users with low Need for Cognition, raising an equity concern about whether cognitive forcing functions widen rather than close the design's effective reach across user populations. The implication for the Veteran Housing Benefit Eligibility Snapshot is that the Snapshot's design must include cognitive forcing functions explicitly (the Assumptions and Unknowns section operates as one) while also recognizing that no single design intervention serves all veterans equally. The hybrid navigator design posture surveyed in Theme 3 addresses this gap directly: digital cognitive forcing functions surface uncertainty for users who can engage with it, and human navigator pathways absorb the load for users who cannot.

Design implications for the Eligibility Snapshot

Across these three sources, three operational design rules for the Snapshot emerge. First, the Snapshot's output structure should and does correspond to the IPDAS-derived decision aid template, which gives the project an established evaluation framework rather than a bespoke one (Stacey et al., 2024). Second, the AI-generated Rationale section is the highest-risk component of the output because well-written explanations reliably amplify trust beyond what the underlying recommendation can sustain (Bansal et al., 2021). The countermeasures the Snapshot deploys are the Assumptions and Unknowns section and the non-determination Disclaimer, which jointly operate as a cognitive forcing function in the Buçinca et al. (2021) sense. Third, the equity finding from Buçinca and colleagues argues for treating the Snapshot as one tier of a multi-tier service in which veterans who need higher-touch support are routed to human navigators rather than left to negotiate the Snapshot's cognitive forcing functions alone.

Theme 2: The Gap Between Eligibility and Access – Algorithmic Harm and Benefits Take-Up

The second theme bounds the design space the Snapshot operates within by surveying two failure modes that the design must avoid in opposite directions. The first failure mode is algorithmic overreach, in which an automated system claims authority over eligibility determinations it cannot reliably make and inflicts concrete harm on vulnerable populations. The second failure mode is the inverse: the eligibility gap that exists when veterans never claim benefits they are entitled to because the application process is too informationally complex or because they were never aware the benefit existed. The two sources in this theme document these two failure modes with empirical specificity, and the Snapshot's design is constrained by both. It must not overclaim, and it must not underdeliver. The space between is the design target.

Eubanks, V. (2018). Automating inequality: How high-tech tools profile, police, and punish the poor. New York, NY: St. Martin's Press.

Eubanks documents three case studies of automated eligibility and decision-making systems deployed against low-income populations in the United States: Indiana's automated benefits eligibility system, the Allegheny County child welfare predictive risk model, and the Los Angeles homeless services coordinated entry algorithm. In each case, the automated system either denied eligibility for benefits the applicant qualified for, prioritized resources in ways that compounded existing disparities, or generated determinations that were treated as authoritative despite being statistically unreliable. The book is journalistic rather than peer-reviewed in venue, but Eubanks is an academic at the University at Albany and the work is widely cited in CHI and CSCW research. The relevance to the Veteran Housing Benefit Eligibility Snapshot tool is direct because the Los Angeles homeless services case in particular involves a partially overlapping population (homeless veterans) and a structurally similar tool (an automated screening system that produces a personalized output). The Eubanks failure mode is what happens when a tool of this class is given authority it does not possess. The Snapshot's design is built explicitly against this failure mode: it produces no determination, makes no prioritization claim, and routes every consequential next step to a human or institutional authority. Whether this design discipline is sustainable as the product scales is a Theme 2 design question; it is not currently a Theme 2 risk.

Bhargava, S., & Manoli, D. (2015). Psychological frictions and the incomplete take-up of social benefits: Evidence from an IRS field experiment. American Economic Review, 105(11), 3489–3529. <https://doi.org/10.1257/aer.20121493>

Bhargava and Manoli conduct a large-scale field experiment with the Internal Revenue Service investigating why tens of thousands of taxpayers fail to claim the Earned Income Tax Credit despite being eligible and having received an initial notice of

eligibility. The experiment varies mailing complexity, program-information framing, and stigma-reduction language across thirty-five thousand non-claimants and measures actual claiming behavior. The headline finding is that simplification of information and heightened salience of the benefit produce substantial additional claiming, while interventions aimed at reducing stigma do not. The authors conclude that low program awareness and informational complexity, not stigma or transaction cost, are the dominant psychological frictions in incomplete benefits take-up. The relevance to Veteran Housing Benefit Eligibility Snapshot tool is foundational. The Snapshot's entire theory of change matches the Bhargava and Manoli finding: the dominant reason eligible veterans do not claim housing-related benefits is informational, and the design intervention with the highest-leverage effect is to reduce the cognitive load of identifying which programs apply and how to act on them. The Snapshot's structured intake form, ranked pathways output, and specific next actions are an operational implementation of the take-up findings from this paper.

Design implications for the Eligibility Snapshot

Eubanks (2018) and Bhargava and Manoli (2015) jointly bound the Snapshot's design space. The Eubanks failure mode argues for the strict non-determination posture: the Snapshot must never claim authority it does not have, regardless of how confident the underlying language model is about a particular pathway. The Bhargava and Manoli finding argues, more affirmatively, that the right intervention for the much larger population of veterans who never claim benefits they qualify for is exactly what the Snapshot is designed to provide: informational simplification, increased benefit salience, and clear pathways to action. Both findings converge on a single design rule: the Snapshot's primary value is informational orientation toward authoritative bodies, not determination of eligibility, and its design discipline must protect that distinction at every level of the output.

Theme 3: Digital Navigation and Plain-Language Design for First-Time Benefits Seekers

The third theme addresses what effective orientation looks like for the Snapshot's specific user population: veterans seeking benefits for the first time after service, typically without prior experience navigating the Veterans Administration or state and local benefits systems. Two peer-reviewed sources anchor this theme. The first is a survey study of user needs and experiences with digital navigation tools, which establishes what users expect from a digital tool of this class and where existing tools fail them. The second is a randomized clinical trial of a structured community health worker navigator intervention, which establishes what effective

human navigation looks like for a partially overlapping population. The pairing maps the design space the Snapshot occupies between the digital floor and the human ceiling.

Stafford, E., Brister, T., Duckworth, K., Rauseo-Ricupero, N., & Lagan, S. (2021). Needs and experiences of users of digital navigation tools for mental health treatment and supportive services: Survey study. JMIR Mental Health, 8(6), e27022. <https://doi.org/10.2196/27022>

Stafford and colleagues survey users of digital navigation tools for mental health and supportive services. The substantive domain is not housing benefits, but the methodological frame and the failure modes documented are directly applicable to any tool of the Snapshot's class. Users reported that the most frequent disappointments with digital navigation tools were outdated or inaccurate resource listings, tools that demanded substantial personal information up front before producing useful output, and tools that produced long undifferentiated lists of programs without helping users distinguish among them. Each failure mode maps to a Snapshot design decision. The Additional Resources section is curated to vetted resources only and surfaces a small ranked set rather than an undifferentiated list. The intake form is structured to capture only the inputs the Snapshot's eligibility logic actually requires. And the no-PII architecture lowers the entry threshold so that veterans can run the Snapshot without supplying identifying information that they may not yet be prepared to disclose. The Stafford findings set a floor of user expectations that the Snapshot's design clears, but they do not validate the design's effectiveness, which is what user testing will measure.

Kangovi, S., Mitra, N., Grande, D., White, M. L., McCollum, S., Sellman, J., Shannon, R. P., & Long, J. A. (2014). Patient-centered community health worker intervention to improve posthospital outcomes: A randomized clinical trial. JAMA Internal Medicine, 174(4), 535–543. <https://doi.org/10.1001/jamainternmed.2013.14327>

Kangovi and colleagues report a randomized trial of a structured community health worker intervention for low-income patients transitioning out of the hospital. The intervention significantly improved primary care follow-up rates, mental health outcomes, and patient activation compared with usual care. The mechanism the authors identify is that the community health workers operated as navigators who oriented patients toward appropriate next steps, helped them understand and plan against barriers, and explicitly did not make decisions for them. The trial is significant for the Snapshot as the strongest available evidence on what effective navigation by a human looks like for the same broad population (low-income, often experiencing housing instability or housing insecurity, frequently first-time interactors with the system in question). The implication is that the Snapshot should be designed to do at scale what a

Kangovi-style community health worker does in person: orient, contextualize, surface document requirements, route toward next steps, and reserve decision-making authority for the institutions that actually possess it. The Snapshot's Next Actions and Document Checklist sections are direct analogues to a community health worker's transition-planning checklist.

Design implications for the Eligibility Snapshot

The Stafford and Kangovi sources jointly support a design posture for the Snapshot that is best characterized as digital community health worker. The Stafford findings identify the floor of user expectations: fresh resource information, minimal data collection up front, and clear next-step routing rather than overwhelming program lists. The Kangovi findings identify the ceiling of human navigator effectiveness and supply the operational template: orient users, surface document requirements, plan against barriers, and refer to decision-making authorities. Plain-language and reading-level discipline are part of this same design posture: regulatory text in the Snapshot's output is rewritten to a sixth-to-eighth grade reading level, acronyms are expanded on first use, and active-voice phrasing is preferred. These conventions are codified in the Federal Plain Language Guidelines maintained by the Plain Language Action and Information Network (referenced contextually, not as a research source) and are operationalized in the Snapshot's output style guide.

Synthesis: Design Implications for the Eligibility Snapshot

The seven sources reviewed jointly support a small, coherent set of design rules for the Veteran Housing Benefit Eligibility Snapshot. The rules are presented in the order in which they affect the user's experience of the output, with supporting sources noted parenthetically.

First, the Snapshot's output structure should and does correspond to an established personalized decision aid template, which provides an evaluation framework grounded in the International Patient Decision Aid Standards rather than a bespoke evaluation invented for this project (Stacey et al., 2024). Second, the AI-generated Rationale section is the highest-risk output component because well-written explanations reliably increase user acceptance beyond what the underlying recommendation can support, and the Assumptions and Unknowns and Disclaimer sections must operate explicitly as cognitive forcing functions to break the rationale's fluency with disclosed uncertainty (Bansal et al., 2021; Buçinca et al., 2021). Third, the Snapshot must never claim eligibility determination authority it does not possess, both as a matter of accuracy and as a defense against the failure mode documented in algorithmic eligibility systems deployed against overlapping populations (Eubanks, 2018). Fourth, the Snapshot's primary value proposition for the much larger population of veterans who never

claim benefits they qualify for is informational simplification and increased benefit salience, both empirically validated as the highest-leverage interventions for incomplete benefits take-up (Bhargava & Manoli, 2015). Fifth, the Snapshot's output must surface a small curated set of next actions and resources rather than an undifferentiated list, must require minimal data collection from the user up front, and must maintain freshness in the resources it surfaces (Stafford et al., 2021). Sixth, the Snapshot is best understood as one node in a hybrid navigator system that includes human escalation pathways for users whose situations exceed what a structured one-shot tool can usefully address, modeled on the demonstrated effectiveness of human community health worker navigators in partially overlapping populations (Kangovi et al., 2014).

Gaps in the Literature

Three gaps in the existing peer-reviewed literature shape the contribution the Snapshot is positioned to make. First, the decision aid literature is mature in clinical domains but sparse in benefits navigation specifically, and the Snapshot's adaptation of the IPDAS framework to housing benefit eligibility is, to the extent surfaced by this review, novel. Second, the benefits take-up literature is dominated by interventions delivered through traditional channels (mailings, in-person assistance, targeted outreach campaigns); the application of take-up theory to an AI-generated personalized report is, again to the extent surfaced by this review, an open research question. Third, the empirical literature on AI overreliance and cognitive forcing functions is primarily set in clinical and decision-support contexts; its application to benefits navigation, where the consequences of overreliance differ in kind from clinical settings, is largely untested. These three gaps do not invalidate the design implications surveyed above; they identify the contribution opportunity for this project. The Snapshot sits at the intersection of three established research traditions and applies their findings to a domain (veteran housing benefit navigation) where none has yet been systematically applied.

References

- Bansal, G., Wu, T., Zhou, J., Fok, R., Nushi, B., Kamar, E., Ribeiro, M. T., & Weld, D. S. (2021). Does the whole exceed its parts? The effect of AI explanations on complementary team performance. In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21). Association for Computing Machinery.
<https://doi.org/10.1145/3411764.3445717>
- Bhargava, S., & Manoli, D. (2015). Psychological frictions and the incomplete take-up of social benefits: Evidence from an IRS field experiment. *American Economic Review*, 105(11), 3489–3529. <https://doi.org/10.1257/aer.20121493>

- Buçinca, Z., Malaya, M. B., & Gajos, K. Z. (2021). To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1), Article 188.
<https://doi.org/10.1145/3449287>
- Eubanks, V. (2018). *Automating inequality: How high-tech tools profile, police, and punish the poor*. New York, NY: St. Martin's Press.
- Kangovi, S., Mitra, N., Grande, D., White, M. L., McCollum, S., Sellman, J., Shannon, R. P., & Long, J. A. (2014). Patient-centered community health worker intervention to improve posthospital outcomes: A randomized clinical trial. *JAMA Internal Medicine*, 174(4), 535–543. <https://doi.org/10.1001/jamainternmed.2013.14327>
- Stacey, D., Lewis, K. B., Smith, M., Carley, M., Volk, R. J., Douglas, E. E., Pacheco-Brousseau, L., Finderup, J., Gunderson, J., Barry, M., Bennett, C. L., Bravo, P., Steffensen, K. D., Gogovor, A., Graham, I. D., Kelly, S. E., Légaré, F., Sondergaard, H., Thomson, R., Trevena, L., & Witteman, H. O. (2024). Decision aids for people facing health treatment or screening decisions. *Cochrane Database of Systematic Reviews*, 1, CD001431.
<https://doi.org/10.1002/14651858.CD001431.pub6>
- Stafford, E., Brister, T., Duckworth, K., Rauseo-Ricupero, N., & Lagan, S. (2021). Needs and experiences of users of digital navigation tools for mental health treatment and supportive services: Survey study. *JMIR Mental Health*, 8(6), e27022.
<https://doi.org/10.2196/27022>